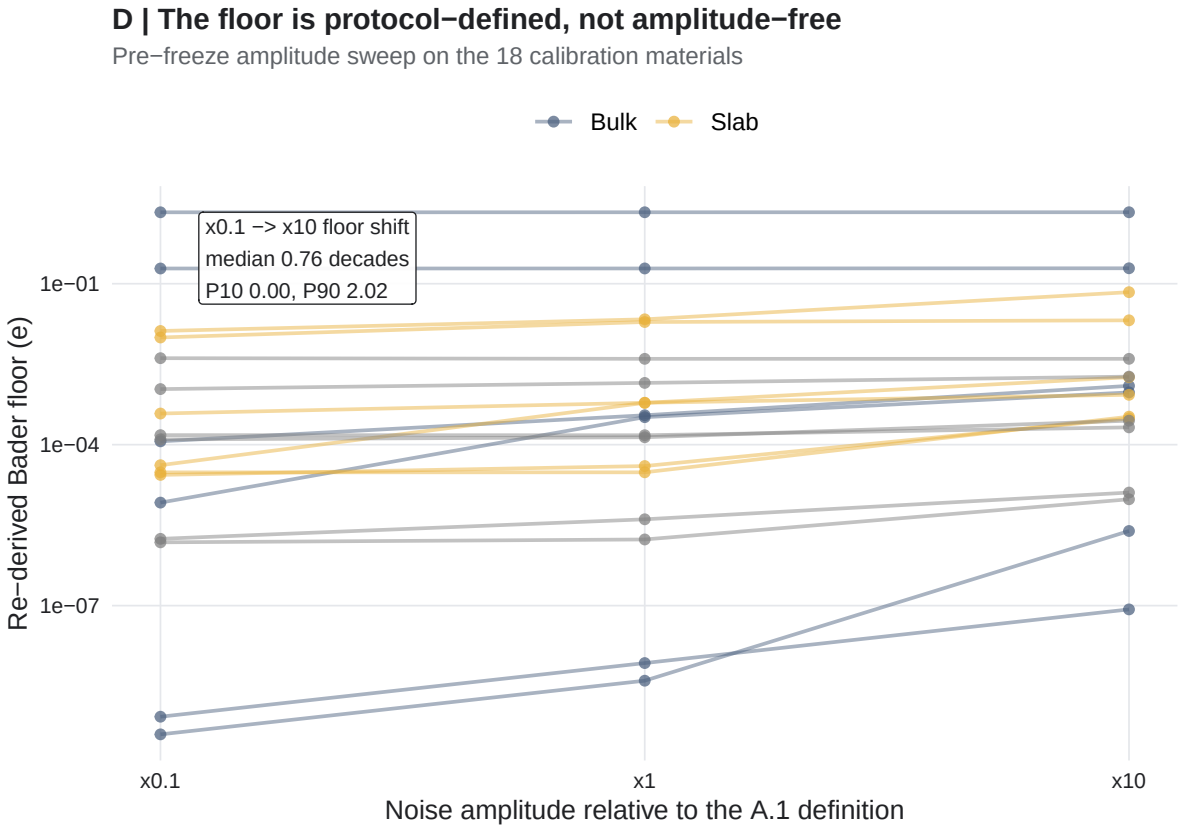
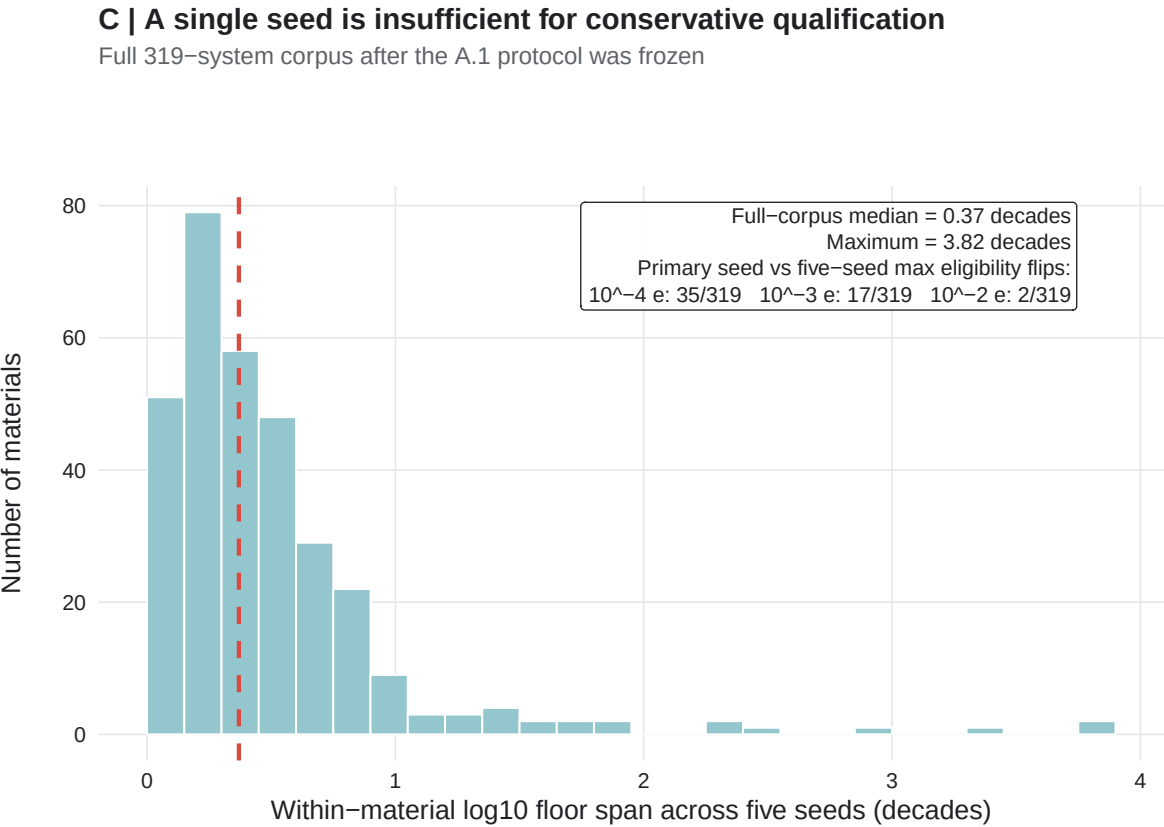
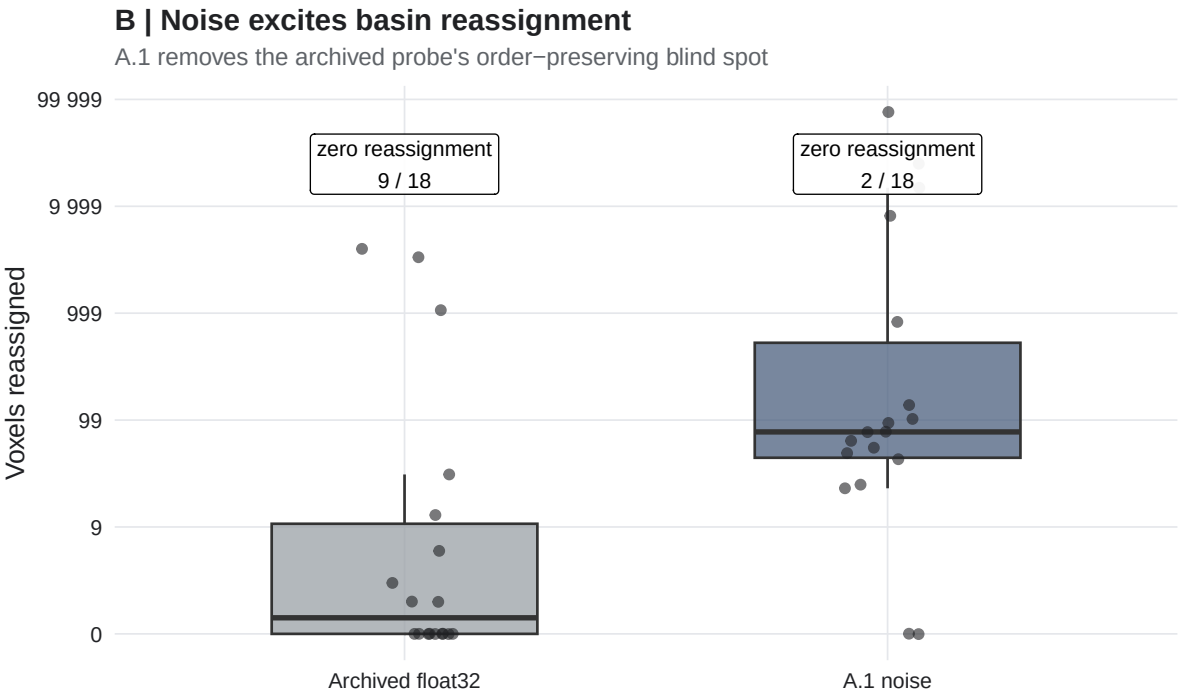
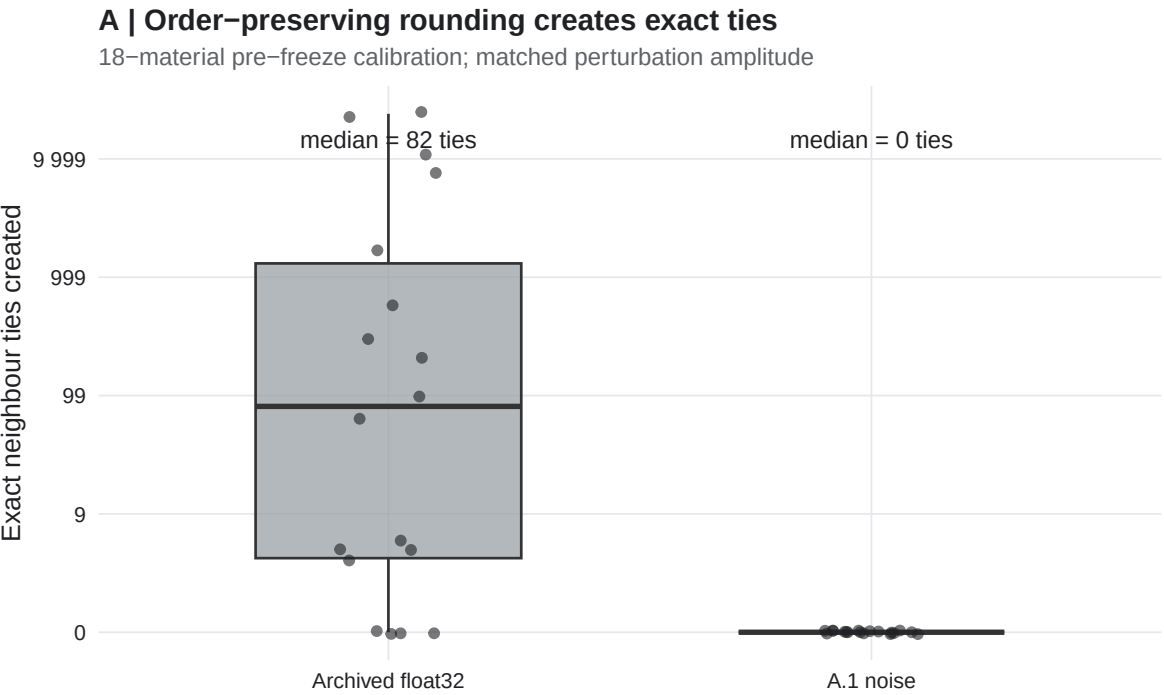


Supplementary Figure S2 | Protocol A.1 probe validation, seed sensitivity and amplitude sensitivity

The calibration justifies a non-order-preserving, multi-seed probe; full-corpus statistics quantify its deployment behavior without retuning.



A–B, 18-material pre-freeze probe calibration. C, five-seed variability in all 319 systems. D, pre-freeze two-decade amplitude sensitivity. The A.1 amplitude and seed set remained frozen.